QP CODE: 24900068

Reg No:.....

Name:.....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

FIRST SEMESTER MGU-UGP (HONOURS) REGULAR EXAMINATION NOVEMBER 2024

First Semester

Discipline Specific Core Course -MG1DSCECC101, MG1DSCCSC100 - ART OF COMPUTING AND PROBLEM SOLVING

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours Maximum Marks: 50

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I), Appreciation (Ap), and Skill (S)

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

[Learning Domain][CO No(s)]

Part A
Very Short Answer Questions
Answer all questions.
Each question carries 1 mark

1	How many bytes are there in a megabyte (MB)?	[U]	[1]
2	Identify the characteristics of a computer that allow it to process large volumes of data quickly.	[U]	[1]
3	Name the error that you encounter if you try to access an array element out of its bounds.	[K]	[1]
4	Name the type of software that is designed to manage and control the hardware components of a computer.	[K]	[1]
5	Which symbol indicates that the line is a preprocessor directive?	[K]	[2]
6	Identify the use of '&&' operator.	[U]	[2]
7	Illustrate how the syntax of a for loop is structured?	[U]	[3]

8	Describe the outcome of the following expression: 5 + 2 * 3.	[U]	[2]
9	How do you declare a string in C?	[K]	[3]
10	How do you skip to the next iteration of a loop?	[U]	[3]

(10 × 1 = 10)

Part B

Short Answer Questions

Answer any 4 questions.

Each question carries 5 marks

11	List the steps of the problem-solving life cycle and briefly explain each step.	[K]	[1]
12	Write an algorithm to find the sum of first n natural numbers.	[U]	[1]
13	Demonstrate the use of at least five keywords in C with an application program.	[U]	[2]
14	Differentiate goto statement and structured control flow statements in C.	[U]	[3]
15	Discuss the commonly used string handling functions in C, highlighting their purpose and usage with examples.	[U]	[3]
16	When would you use the break statement in loops? Explain with an example.	[U]	[3]

(4 × 5 = 20)

Part C

Essay Questions

Answer any 2 questions.

Each question carries 10 marks

17	Explain the differences between syntax errors, logical errors, and runtime errors. Provide examples for each type of error and how they can be resolved.	[U]	[1]
18	Discuss the use of variables and constants in programming. Explain different types of constants with examples.	[K]	[2]
19	Explain the different forms of 'if' statements with syntax and examples.	[U]	[3]

(2 × 10 = 20)

END OF THE QUESTION PAPER

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